

1. 2012Q22

Which of the following descriptions about the function of the cilia lining the trachea is correct?

- A. to warm the incoming air
- B. to moisten the incoming air
- C. to move mucus away from the trachea
- D. to produce mucus to protect the trachea

2. 2013Q1

Question 1 refers to the table below, which shows the relative amounts of mitochondria, chloroplasts and endoplasmic reticulum in four cell types P, Q, R and S:

| Cell type | Relative amount of the sub-cellular structure |             |                       |
|-----------|-----------------------------------------------|-------------|-----------------------|
|           | Mitochondrion                                 | Chloroplast | Endoplasmic reticulum |
| P         | +                                             | ++          | +                     |
| Q         | +++                                           | -           | +                     |
| R         | +++                                           | -           | +++                   |
| S         | +                                             | -           | +                     |

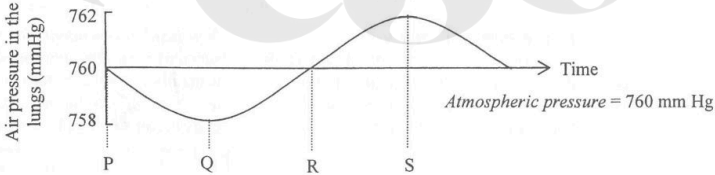
Key: number of '+' indicates the relative amount of the sub-cellular structure  
'-' indicates the absence of the sub-cellular structure

Which cell type is found in the lining of the air sacs in the lungs?

- A. P
- B. Q
- C. R
- D. S

3. 2013Q30

The graph below shows the change in air pressure in the lungs of a person:

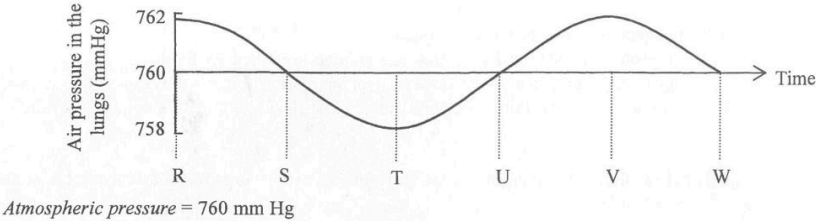


At which point of time on the graph is the volume of the lung the largest?

- A. P
- B. Q
- C. R
- D. S

4. 2014Q27

The graph below shows the change in air pressure in the lungs of a man:

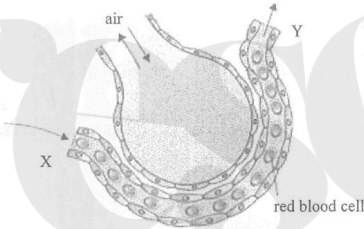


His diaphragm muscles are in a state of contraction during the period

- A. RT
- B. SU
- C. TV
- D. UW

5. 2015Q13

Question 13 refers to the diagram below, which shows a section of an air sac and its associated blood capillary in humans:

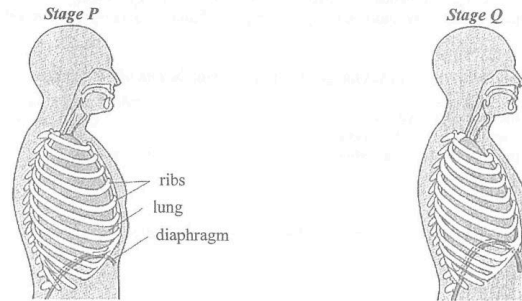


Which of the following combinations correctly describes the changes in blood composition when blood flows from X to Y?

|    | Oxygen content    | Glucose content   | Urea content      |
|----|-------------------|-------------------|-------------------|
| A. | increases         | remains unchanged | increases         |
| B. | increases         | decreases         | remains unchanged |
| C. | remains unchanged | decreases         | remains unchanged |
| D. | remains unchanged | remains unchanged | increases         |

6. 2015Q15

The diagram below show the relative positions of the human respiratory system and its associated structures in two different breathing stages:

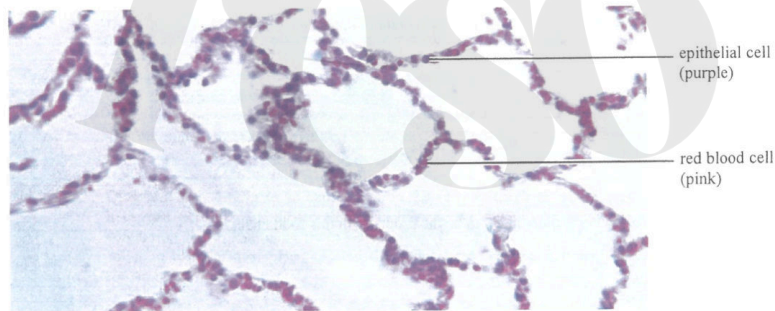


Which of the following statements correctly describes the change that takes place from stage P to stage Q?

- A. Pressure inside the lungs is increasing.
- B. Diaphragm muscle is contracting.
- C. Volume of the lungs is increasing.
- D. Rib cage is moving upward.

7. 2018Q8

The photomicrograph below shows a section of human lung:



With reference to the structures shown in the photomicrograph, which of the following are adaptive features for gas exchange?

- (1) presence of water film
- (2) short diffusion distance
- (3) rich supply of blood capillaries

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

8. 2020Q11

Which of the following descriptions of breathing is correct?

- A. The air rushes in to expand the lungs.
- B. The trachea expands to draw in more air.
- C. Pressure in the lungs decreases to draw in air.
- D. The expansion of the lungs pushes the diaphragm downward.

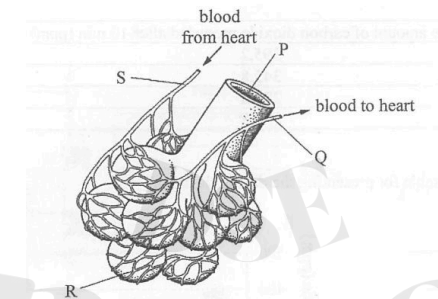
9. 2020Q13

Both the villi of the small intestine and the air sacs of the lung are sites for material exchange. Which of the following is an adaptive feature common to the epithelium of both structures?

- A. The epithelium is one cell thick.
- B. There is a water film on the surface of the epithelium.
- C. The epithelial cells contain a large number of mitochondria.
- D. The epithelial cells have a modified cell membrane to increase surface area.

10. 2022Q15

Questions 15 and 16 refer to the diagram below, which shows the end of the respiratory tract and the associated blood vessels in humans:



Which labelled part has the highest concentration of carbon dioxide?

- A. P
- B. Q
- C. R
- D. S

11. 2022Q16

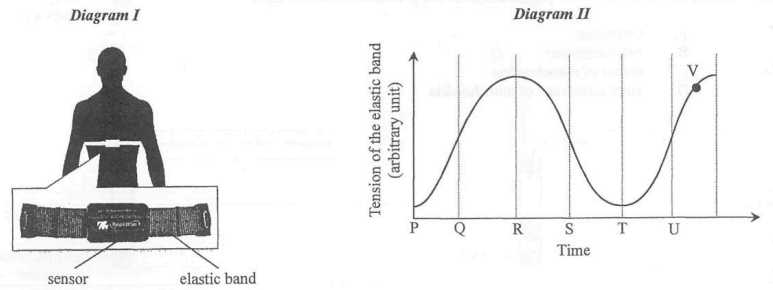
Which of the following adaptive features for gas exchange can be illustrated in the above diagram?

- (1) The respiratory tract ends in numerous spherical structures.
- (2) The end of the respiratory tract has a thin wall.
- (3) The end of the respiratory tract is covered with blood vessels.

- A. (1) and (2) only
- B. (2) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

12. 2022Q17

Questions 17 and 18 refer to the diagrams below. Diagram I shows a device which is used to monitor the breathing pattern of a person. It consists of an elastic band strapped around a person's chest and a sensor which detects the tension of the elastic band. Diagram II shows the change in the tension of the elastic band during breathing.



Exhalation takes place during period

- A. PR.
- B. QS.
- C. RT.
- D. SU.

13. 2022Q18

Which of the following interpretations about Diagram II is/are correct?

- (1) The diaphragm is contracting during period QR.
  - (2) Air pressure inside the lungs increases during period TU.
  - (3) Air pressure inside the lungs is greater than atmospheric pressure at point V.
- A. (1) only
  - B. (2) only
  - C. (1) and (3) only
  - D. (2) and (3) only

14. 2023Q9

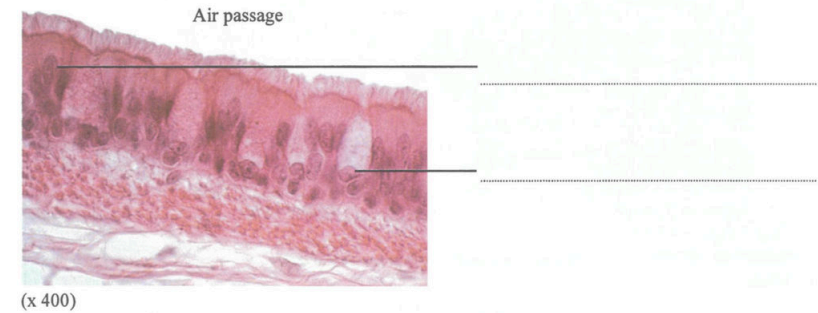
When we breathe out, some air will be left inside the air sacs in the lungs. Which of the following correctly shows the gas composition of the remaining air inside the air sacs?

|    | Oxygen | Carbon dioxide |
|----|--------|----------------|
| A. | 16%    | 0.04%          |
| B. | 16%    | 4%             |
| C. | 21%    | 0.04%          |
| D. | 21%    | 4%             |

LQ

1. 2014Q2a

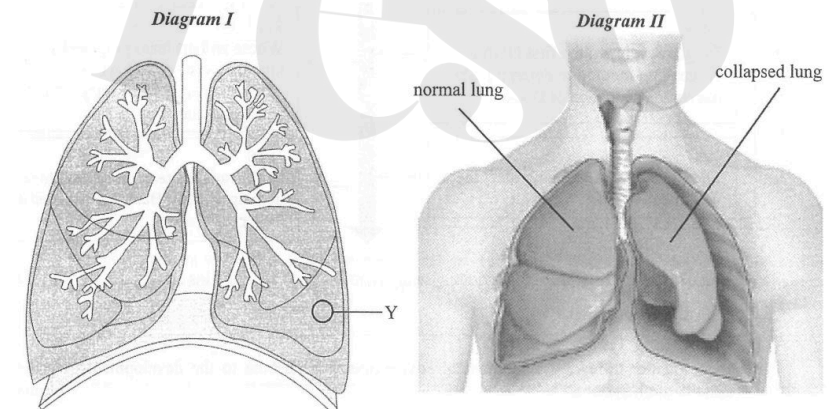
The photomicrograph below shows a section of the inner wall of the human trachea:



- a) In the spaces provided, label the cells shown in the photomicrograph. (2 marks)

2. 2017Q5

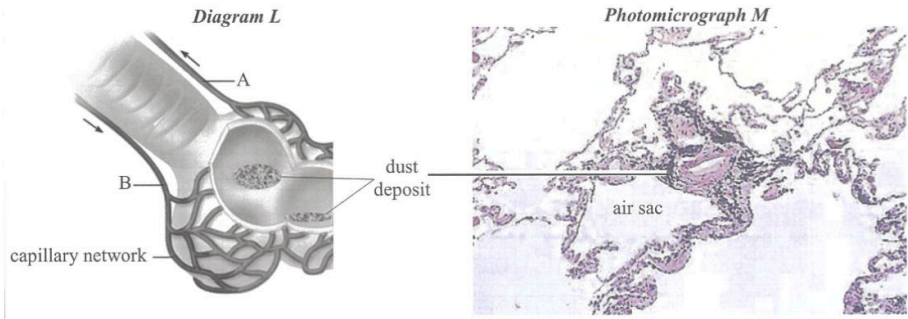
- a) Briefly describe the breathing actions that bring air into the lungs. (4 marks)
- b) In the following diagrams, Diagram I shows some structures of a human lung while Diagram II shows a collapsed lung if it is ruptured at location Y:



Explain why the lung collapses if it is ruptured at location Y.

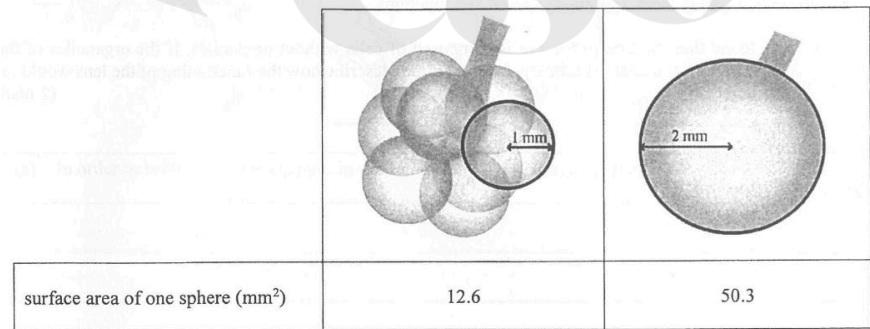
(2 marks)

3. 2019Q5  
Diagram L below shows part of the lung in a patient suffering from a certain lung disease. A hardened layer of dust deposit was found on the respiratory surface of the air sacs. Photomicrograph M shows the lung tissue taken from the patient.



- a) Compare the oxygen and glucose content of the blood in vessels A and B. Explain your answer. (4 marks)
- b) With reference to the above information about the lung disease, suggest **two** possible ways in which the disease adversely affects gas exchange in the patient. (4 marks)

4. 2023Q1  
The spheres shown in the diagram below represent the air sacs of different sizes in the lung. The total volume of the eight spheres with a radius of 1 mm each is equal to the volume of one large sphere with a radius of 2 mm.



- a) Calculate the total surface area of eight small spheres. (1 mark)
- b) With reference to the answer in (a), explain why having smaller air sacs in the lungs is more efficient than bigger air sacs for gas exchange. (2 marks)
- c) Apart from (b), explain how air sacs are specialised at tissue level for gas exchange. (1 mark)

Ans

|                  |                  |                  |                  |
|------------------|------------------|------------------|------------------|
| 2012Q22: C (84%) | 2013Q1: D (54%)  | 2013Q30: C (7%)  | 2014Q27: B (62%) |
| 2015Q13: B (73%) | 2015Q15: A (76%) | 2018Q8: C (63%)  | 2020Q11: C (80%) |
| 2020Q13: A (82%) | 2022Q15: D (70%) | 2022Q16: B (48%) | 2022Q17: C (69%) |
| 2022Q18: A (61%) | 2023Q9: B (44%)  |                  |                  |

- 2014Q2a  
2 (a) ciliated epithelial cell (1)  
mucus-secreting cell (1) (2)

- 2017Q5  
5 (a) • intercostal muscle contracts to result in upward and outward movement of rib cage / raise the rib cage (1)  
• diaphragm muscle contracts to flatten the diaphragm (1)  
• both actions increase the volume of the thoracic cavity (1)  
• as a result, the pressure inside the lungs drops below atmospheric pressure (1)  
Hence air is forced into the lungs (4)
- (b) • air will leak into the pleural cavity (1) from outside through the lungs / the negative pressure of pleural cavity cannot be maintained (2)  
• the lung collapses due to its own elasticity (1) (2)
- 6 marks

- 2019Q5  
5 (a) • A contains more oxygen than B / B contains less oxygen than A (1)  
because gas exchange takes place at air sacs where oxygen is taken up by blood through diffusion (1), so blood leaving air sac should have more oxygen (4)  
• A contains less glucose than B / B contains more glucose than A (1)  
because the cells in air sacs / lung tissues have taken up glucose from the blood for respiration (1), so blood leaving air sac should have less glucose
- (b) Any **two** sets:  
• dust deposits form a barrier (1), which increases the diffusion distance (1)  
• the inner surface of the air sacs was covered by dust(1), thus the area available for diffusion was reduced (1)  
• the hardened layer of dust deposit reduces the elasticity of the lung (1), hence there is a smaller lung volume when the lung inflates (1) (4)
- 8 marks

- 2023Q1  
1 (a) 100.8mm<sup>2</sup> (1)  
[no unit, no mark] (1)
- (b) Concept for mark award:  
• correct comparison of the surface are / surface area to volume ratio (1)  
• significance of the difference to the diffusion of gases (1) (2)
- e.g.  
• for the same volume, 8 small spheres have a larger total surface area / a larger surface area to volume ratio than a large sphere (1)  
• therefore, the diffusion rate for gas exchange will be higher / diffusion of gases will be faster (1) (1)
- (c) any **one** of the following:  
• thin wall (made up of a single layer of cells) offers a short diffusion distance for rapid gas exchange (1)  
• some cells in the wall secrete fluid / the presence of a water film on the inner wall for gas dissolution / allowing gases to dissolve (1) (1)
- 4 marks